Multi-Stage Transfer Learning for Efficient Video Prediction and Segmentation with Sparse Labeled Data

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Abstract

Video prediction and segmentation are important tasks in computer vision. They usually require large amounts of high-quality labeled data to achieve good performance, which is expensive and time-consuming in real-world scenarios. In this work, we propose a multi-stage transfer learning algorithm along with a purely CNN-based model that effectively addresses the challenge of scarce labeled data for end-to-end video prediction and segmentation tasks in a unified framework. Our model achieves a Jaccard Index of 0.4025 on the hidden test set, ranking 3rd among 31 teams in a graduate-level deep learning course supervised by Professor Yann LeCun and Professor Alfredo Canziani.

1 Introduction

Computer vision applications, such as autonomous vehicles and surveillance systems, heavily rely on video prediction and segmentation for accurate scene understanding and decision-making. The performance of these tasks heavily relies on large amounts of high-quality labeled data. However, obtaining such data can be expensive and time-consuming in real-world scenarios, posing significant challenges for researchers and practitioners.

Our contribution mainly lies in two aspects. First, we propose a novel multi-stage transfer learning algorithm, along with a purely CNN-based model, that effectively addresses the challenge of scarce labeled data for video prediction and segmentation tasks. Our approach focuses on leveraging available labeled data and enriching the dataset to facilitate the training of more complex models while maintaining high performance in both prediction and segmentation tasks.

The second key innovations of our method is the integration of end-to-end prediction and segmentation into a single framework. This enables our model to learn the spatiotemporal features of video data more efficiently and perform both tasks simultaneously. By incorporating end-to-end prediction and segmentation, we not only improve the efficiency of our method but also enhance its generalization capabilities, making it more robust and applicable to a wider range of scenarios.

2 Related works

Transfer learning in computer vision Transfer learning is a popular technique in computer vision that leverages pre-trained models to solve new tasks with limited data. It has become an essential aspect of computer vision in recent years, significantly improving the performance of various vision tasks by leveraging pre-trained models and knowledge from related domains Pan and Yang [2010]. The models are often fine-tuned on target tasks, resulting in considerable performance improvements.

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Video prediction and segmentation Video prediction and segmentation are two essential tasks in the field of computer vision, enabling applications such as video editing, tracking, and compression. These tasks have attracted significant research attention over the past few years, leading to the development of various state-of-the-art models.

One of the most notable architectures in the field of semantic segmentation is the UNet Ronneberger et al. [2015], which is characterized by its symmetric encoder-decoder structure and skip connections. The UNet has been successfully applied to a wide range of segmentation tasks.

SimVP (Simple Spatiotemporal Video Prediction) by Gao et al. [2022] Tan et al. [2022] is a straightforward baseline model designed for video prediction tasks. Built entirely upon convolutional networks without the need for recurrent architectures, SimVP is trained using a standard mean squared error loss function. Remarkably, this model achieves state-of-the-art results, without the inclusion of any intricate modules, strategies, or techniques, showcasing its efficacy and simplicity.

Pseudo-labeling First introduced by Lee in 2013 Lee [2013], the pseudo-labeling method is an innovative semi-supervised learning technique that leverages a small set of labeled data in conjunction with a vast pool of unlabeled data to enhance a model's performance. This approach addresses the challenge of limited labeled data, which is a common issue in many real-world applications.

3 Methodology

In this section, we will explain the problem setup of the project and the details of the Multi-Stage Transfer Learning algorithm we applied to solve the problem of scarcity of labeled data.

3.1 Problem setting

In this project, we are working with a dataset of video sequences, each consisting of 22 frames. Our objective is to predict the segmentation mask of the 22nd frame, given the first 11 frames as input. The dataset consists of 3 parts as follows.

- Labeled training set: 1000 video sequences with 22 frames of shape 160×240. The segmentation masks of each frame are included in each data point. $\{X_{L,(i,j)}\}_{j\in[22]}$ is the ith labeled training video sequence and $\{M_{L,(i,j)}\}_{j\in[22]}$ is the corresponding segmentation masks.
- Unabeled training set: 13000 video sequences with 22 frames of shape 160×240. The segmentation masks of each frame are not included in each data point. The unlabeled training set is denoted by $\{X_{U,(i,j)}\}_{i\in[13000],j\in[22]}$.
- Validation set: 1000 video sequences with 22 frames of shape 160×240. The segmentation masks of each frame are included in each data point. The validation set is denoted by $\{X_{V,(i,j)}\}_{i\in[1000],j\in[22]}$, $\{M_{V,(i,j)}\}_{i\in[1000],j\in[22]}$.

The segmentation mask for each frame is of shape 160×240, with each entry indicating the class of the pixel, taking values in $\{0, 1, \ldots, 48\}$. (There are 48 different kinds of objects with unique (color, materials, shape) settings. Besides these objects, a pixel might also belong the background.)

3.2 Multi-Stage Transfer Learning

To achieve the objective above, we need to develop a method that can:

- Efficiently learn the spatiotemporal features of the input video sequence (that is, the first 11 frames) to simultaneously predict the segmentation mask of the 22nd frame (that is, an end-to-end fashion).
- Leverage the limited labeled data (i.e., the subset of video sequences with segmentation masks for each frame) effectively, while also utilizing the large amount of unlabeled data to improve the model's performance.

There are 2 steps in our proposed **Multi-Stage Transfer Learning** algorithm for end-to-end segmentation prediction.

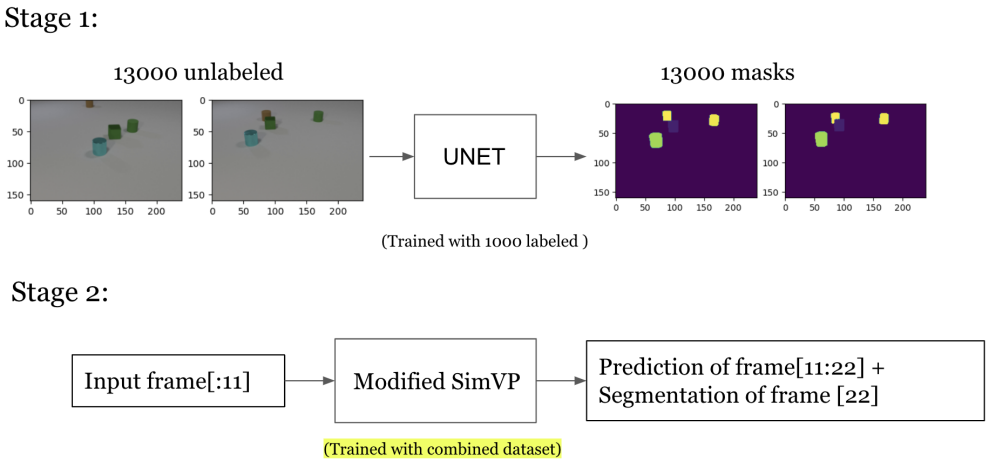


Figure 1: Visualization of our multi-stage transfer learning algorithm.

3.2.1 Step 1: pseudo-labeling

The first step is to train a segmentation model (here we use the U-Net and we believe a more powerful model can improve the performance of our algorithm) using the 1000 labeled video sequences. This U-Net model is then utilized to generate pseudo-labels (predicted segmentation masks) for the 13000 unlabeled video sequences. By transferring the knowledge gained from the U-Net model, we obtain a combined dataset of 14000 sequences, including both original labeled data and pseudo-labeled data.

These pseudo-labels serve as "noisy" or "approximate" ground truth annotations for the unlabeled data. Its quality highly depends on the performance of the segmentation model we choose. Since the video frames in this project are simple shapes, the U-Net trained with 1000 labeled sample can generalize well on the unseen frames. We will report how this pseudo-labeling technique improve the performance of our model in section 4.

3.2.2 Step 2: modified SimVP

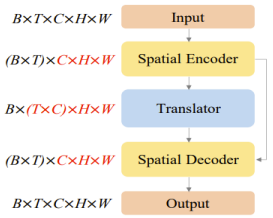


Figure 2: Detailed flowchart of SimVP Gao et al. [2022]

This enriched dataset allows us to train a large CNN-based model called SimVP, which is a state-of-the-art algorithm for video prediction (Gao et al. [2022]). From Figure 2 SimVP consists of an encoder, translator, and a decoder. All these three parts of the model consists of CNN layers (and Normalization layers etc, but no RNN or Transformer included).

SimVP for predicting Segmentation map SimVP is designed for predicting the ground truth frames where each frame is a 3-channel RGB image. In our cases, the objective is to predict segmentation maps of the last 11 frames. As mentioned in Section 3.1, each pixel might belong to 49 different classes. So we now increase the number of CNN kernels from 3 to 49 and for each pixel, we will get 49 logits values. We minimize cross entropy loss on these 49 logits values with labelled segmentation maps or pseudo-labelled segmentation maps.

4 Experiments

In our project, we employ weighted cross entropy as the loss function to train the U-Net model as well as the segmentation part of SimVP. Weighted cross entropy assigns different weights to each class, effectively lowering the weight of the background class in the loss calculation. This modification ensures that the model pays more attention to the object classes during training, rather than being biased towards the more prevalent background class.

We have compared performance of three different (modified) SimVP training strategies:

1. Train SimVP end-to-end on (pseudo-labelled and labelled) segmentation maps of combined 14000 sequences
2. Pretrain SimVP on 14000 ground truth video sequences and finetune the models on segmentation maps of 1000 labelled video sequences
3. Train SimVP on 14000 ground truth video sequences and use segmentation UNet to transform SimVP predictions into segmentation maps.

We evaluate model f on validation set by:

1. Jaccard index and MAE between predicted segmentation of last 11 frames $f(\{X_{V,(i,j)}\}_{j\in[12,22]})$ and the labelled segmetation of the corresponding frames $\{M_{V,(i,j)}\}_{j\in[12,22]}$ for $i \in [1000]$.
2. Jaccard index and MAE between predicted segmentation of last frame $f(\{X_{V,(i,22)}\})$ and the labelled segmetation of the corresponding frame $\{M_{V,(i,22)}\}$ for $i \in [1000]$.

	End-to-End	Pretrain and Fintune	Two Stage
Jaccard Index(last frame) ↑	**0.4165**	0.2870	0.2261
Jaccard Index(11 frames)↑	**0.5829**	0.4170	0.2856
MAE(last frame) ↓	**0.9401**	1.0323	1.4977
MAE(11 frames) ↓	**0.6422**	0.7855	1.2365

Table 1: Performance of SimVP models trained with different strategies

From Table 1 we can see that SimVP trained end-to-end on pseudo-labelled dataset performs better than other two training strategies on both predicting the last one frame in 22-frame sequence and the last 11 frames in 22-frame sequence. One example of predicted segmentation of the last frame given by training an end-to-end SimVP on pseudo-labeling is recorded in Figure 3. More examples are available at AppendixA.



Figure 3: ground truth last frame(left); labeled segmentation of the last frame (centered); predicted segmentation of the last frame (right)

5 Conclusion

In this research we develop an end-to-end training strategy to train modified SimVP on video prediction and segmentation task. Moreover, we employ pseudo labeling trick to fully utilize the potential of a sparsely labeled dataset. Our experiments reveal that an end-to-end models trained with pseudo-labelling trick provides more promising performance on video prediction and segmentation task than non end-to-end model and end-to-end model trained without the use of pseudo-labeling.

References

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A More Examples of predicted segmentation maps

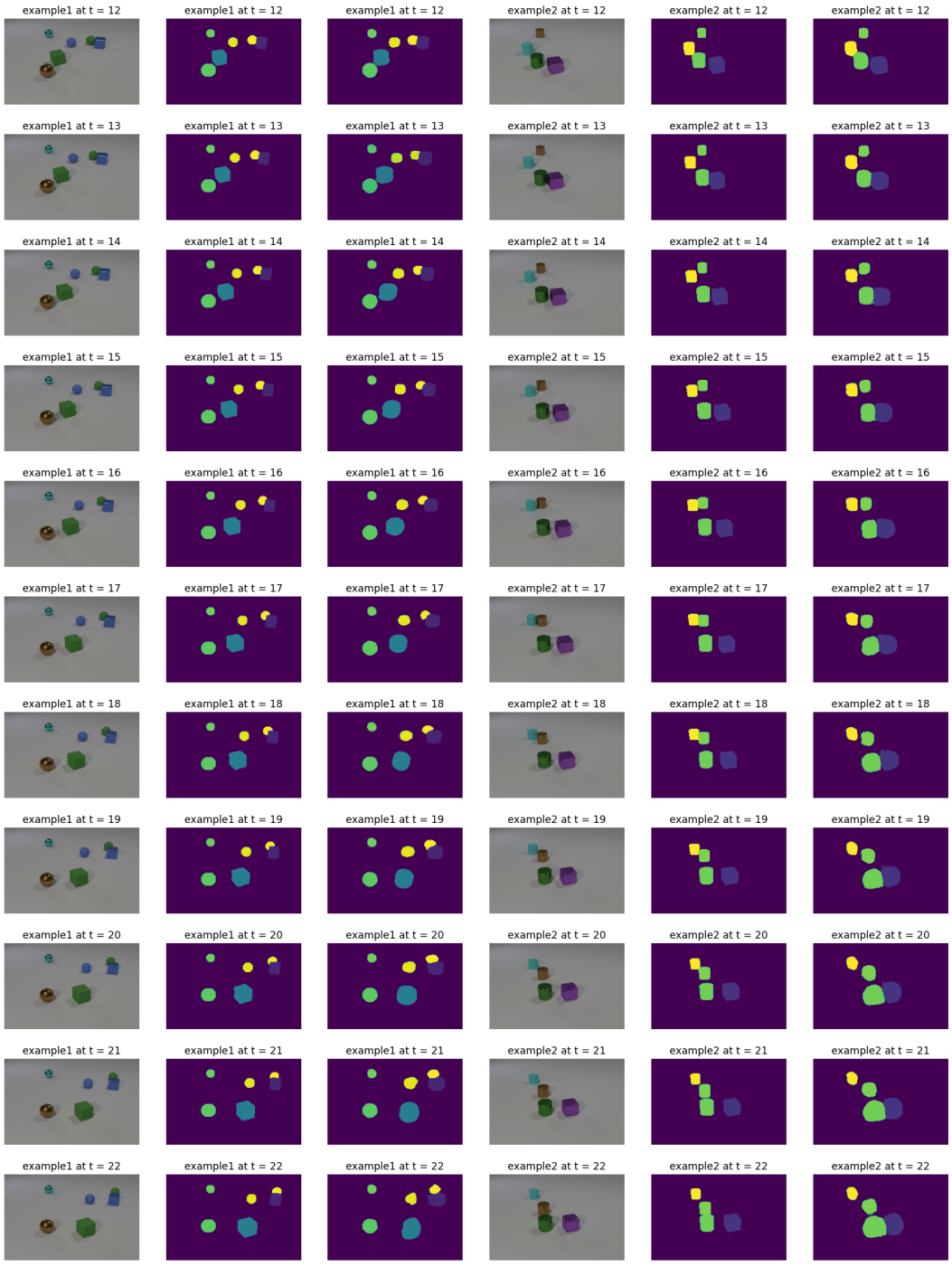


Figure 4: ground truth image(left), labeled segmentation maps (center); predicted segmentation maps (right)

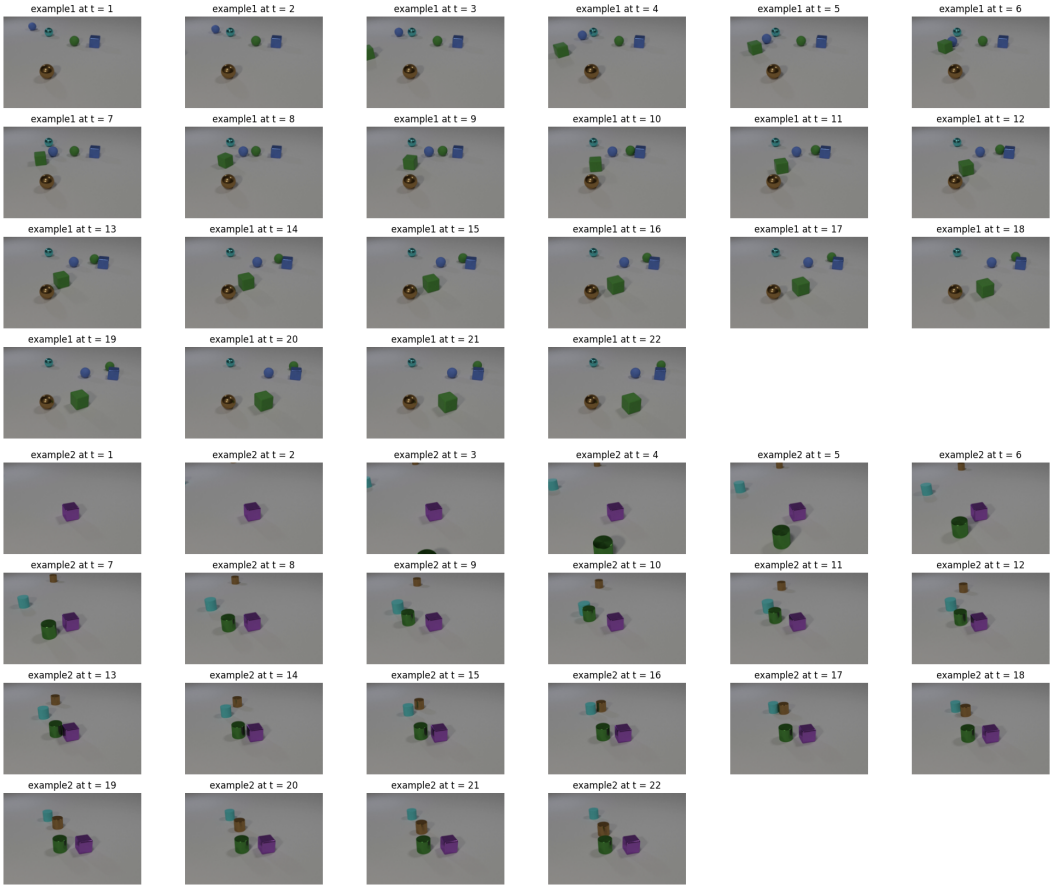


Figure 5: the complete 22-frame video sequence for the above two examples